

WASP-11b

R-0746

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_231220 · created 2026-07-28T02:22:48+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460298.7419 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.052
Transit depth (Rp/Rs)^2	3.09 +/- 1.82 [%]
Orbital Inclination (inc)	88.7 +/- 1.67
Airmass coefficient 1 (a1)	2.92 +/- 1.76
Airmass coefficient 2 (a2)	-0.95 +/- 0.82
Scatter in the residuals of the lightcurve fit is	61.86 %
Best Comparison Star	#2 - [517, 227]
Optimal Aperture	3.22
Optimal Annulus	8.82
Transit Duration (day)	0.102 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.176 ± 0.052 vs 0.131 (deviation 0.87σ)
Duration fit vs published	0.102 d vs 0.1075 d (deviation 0.34σ)
Mid-time O–C	7.9 min
Depth SNR	3.4
Residual scatter	61.86 %

Light curve

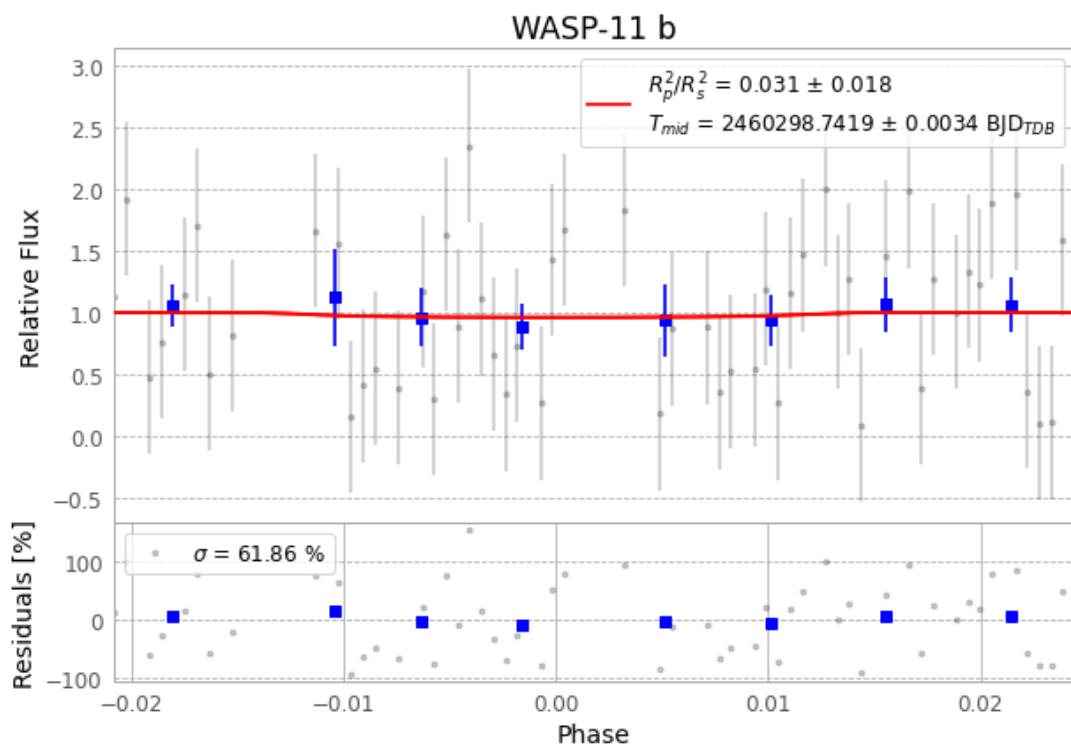


Figure 1 — FinalLightCurve_WASP-11 b_2023-12-19.png (SHA-256 d9d99f4fdd6aaebf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [394, 281], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 06938d526a32ca9e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

82 FITS frames (54 MB), HATP-10231220034816.FITS → HATP-10231220075115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-231220.log (b48ab667b7d6...), FinalLightCurve_WASP-11 b_2023-12-19.png (d9d99f4fdd6a...), FinalLightCurve_WASP-11 b_2023-12-19.csv (60ae24c33217...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 02:22Z.